ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = 5 V/2^{12} = 5/4096 = 1.2207 mV = 1.2207×10^{-3} V.

ΔF = LSB/S with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) $F = e^{AT}$, $G = A^{-1}(F - I)B$. Taking T = 0.01 s.

$F = e^{-0.02} =$ **0.9802**.

$G = (0.9802 - 1)/(-2) =$ **0.0099**.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\,x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Since $|\sin\theta| \le 1$, an equilibrium exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) Jacobian at $\theta_e = 60°$: A = [0, 1; $-(mg\ell/J)\cos\theta_e$, $-b/J$].

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss} = 2.4525 \sin\theta_{ss}$. At $\theta_d = 90°$ the right side ≠ 0 while the left side = 0, so the joint cannot rest at θ_d.

With $\sin\theta_{ss} \approx 1$: $\theta_d - \theta_{ss} = 2.4525/50 = 0.00491$ rad = **0.281°**.

(d) With the compensation term, at rest $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss}$, so $\theta_{ss} = \theta_d$ and the steady-state error is **0**.

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\, r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$.

With a = 2, b = 1, c = 6, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain = $bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

(b) With b = 0.75 ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} =$ **0.8**. Tracking error = (1 − 0.8)/1 × 100 = **20%**. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) Substituting u into the plant and using $\dot{\xi} = -\theta$ (r = 0): characteristic polynomial $s^3 + (a + k_2)s^2 + (c + k_1)s + K_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 24 \Rightarrow$ **$k_1 = 18$**; **$K_I = 32$**.

(d) Becuase the closed loop is stable, the steady-state error is zero for any b and c.

Failure mode: integrator windup. Remidy: clamp the integrator output (anti-windup).